

CIS 206 - Assignment 2 Coding Standard

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I have never written a document on code like this before, so I am unfamiliar with how such a document should be formatted, but I will try my best. In the following texts, I will break down my personal standards, thought process and standards I learned through research.

I was not in a group for this assignment, and did everything on my own.

Variables

When handling variables, I typically try to stay away from using them from having multiple inputs, meaning that I create them to handle one source of data. With this in mind, my variable names are often hyperspecific and named after what I will be using them for.

weight is a very straightforward value that is filled via user input. Since nothing else needs to be done with this variable, it is only used in other equations.

inch and **feet** are both inputs fed by the user and later added together to create the value for **height**.

height; as mentioned is the combined values of **inch** and **feet**. Due to how those units are handled, **feet** is broken down into inches by multiplying its value by 12, then the **inch** value is added on at the end.

bmi is our numeric value for the user's Body Mass Index (BMI). The equation for this utilizes **weight** and **height**, the formula being: $\text{weight}/\text{height}^2$.

report is a variable that is filled with text based on the user's BMI value after it runs through the **report_process** function.

Functions

Similar to my variables, I don't tend to like overcrowding functions. Functions are meant to be repeatable actions, but that does not mean that the entire program should be nested inside one

alone. In my case, each type of processing variable is handled in a separate function, often named after what the exact purpose of the function is.

height_process is the function that handles the actual formula that generates height. Taking feet and inch as input variables, it runs the equation: **height = (feet * 12) + inch** and returns **height** back to *main*.

bmi_process is very similar, taking in **weight** and **height** as input variables, running the formula: **bmi = (weight/height** 2)** and returning **bmi** to *main*.

report_process differs from the two above due to the fact it does not return a numeric value like the others. Rather, this function has if-else statements that check the user's **BMI** value, then returns a text string into the **report** variable.

Other Standards

In addition to everything above, there are other standards I have not listed just yet. When it comes to writing names for variables or functions, I am used to doing camel case (**formattedLikeThis**) in other languages, but in Python I keep everything lowercase and broken up with underscores (**formatted_like_this**). This makes names of items readable and compatible with the language.

Something I recently learned for Python was limiting characters per line to 79 characters to avoid text wrapping. Subconsciously this is something I always do my best to avoid, even when writing functions. The only real exception I have to this rule is when pasting a link to documentation (as seen in this program and WHO's website for BMI).

In conclusion, my standards are to keep things readable not only for myself, but for others who may look at the code at some point in the future. Even if a viewer does not know how to code, they should be able to read comments and variable/function names to get a general understanding of what is going on.